

Section 4I. Univariate LDA

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Univariate LDA

- **Univariate case** ($p = 1$):

- For $p = 1$, the Gaussian density has the form

$$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} e^{-\frac{1}{2}\left(\frac{x-\mu_k}{\sigma_k}\right)^2}$$

- Assuming identical variances $\sigma_k = \sigma$, we have that

$$\begin{aligned} \arg \max_k f_k(x) \pi_k &= \arg \max_k \log(f_k(x) \pi_k) = \dots \\ &= \arg \max_k \left(\frac{\mu_k x}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log \pi_k \right) = \arg \max_k \delta_k(x) \end{aligned}$$

where $\delta_k(x)$ is a linear function of x

- In practice, we are not explicitly given μ_k , σ , and π_k , but a dataset...

Univariate LDA (cont.)

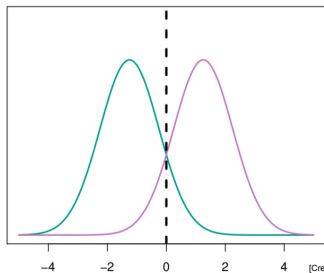
- We can estimate the parameters μ_k , σ , and π_k from a dataset, using the following formulas:

$$\begin{aligned}\hat{\pi}_k &= \frac{N_k}{N} \\ \hat{\mu}_k &= \frac{1}{N_k} \sum_{i: y_i=k} x_i \\ \hat{\sigma}^2 &= \frac{1}{N-K} \sum_{k=1}^K \sum_{i: y_i=k} (x_i - \hat{\mu}_k)^2 \\ &= \sum_{k=1}^K \frac{N_k - 1}{N - K} \hat{\sigma}_k^2\end{aligned}$$

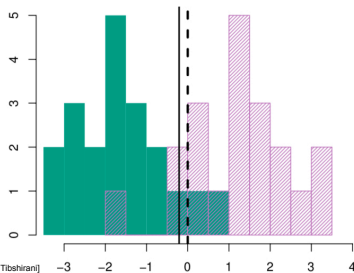
where N_k is the number of samples in the class k and $\hat{\sigma}_k^2 = \frac{1}{N_k - 1} \sum_{i: y_i=k} (x_i - \hat{\mu}_k)^2$

Univariate LDA Example

Consider two Gaussians with $\mu_1 = -1.5$, $\mu_2 = 1.5$, $\pi_1 = \pi_2 = 0.5$ and $\sigma^2 = 1$



[Credit: Hastie and Tibshirani]

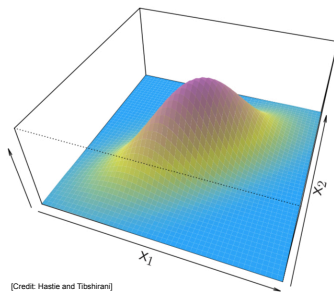


Multivariate LDA

- Multivariate LDA ($p > 1$): The Gaussian density takes the form

$$f_k(\mathbf{x}) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} e^{-\frac{1}{2}(\mathbf{x}-\mu_k)^T \Sigma^{-1}(\mathbf{x}-\mu_k)}$$

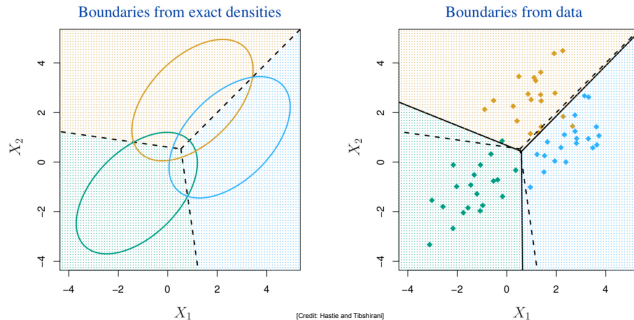
where $\mathbf{x}$ and μ_k are p -dimensional vectors and Σ is a $p \times p$ matrix (and $|\Sigma|$ is its determinant)

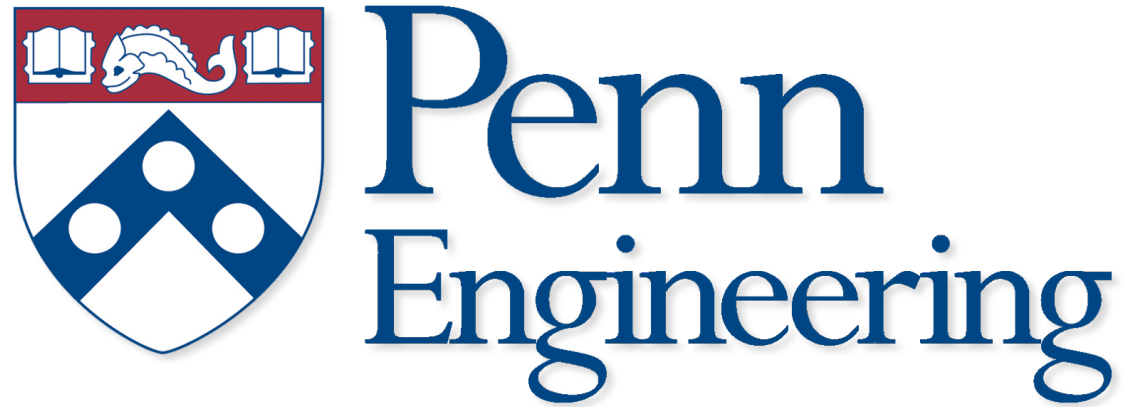


[Credit: Hastie and Tibshirani]

Multivariate LDA: Numerical example

- Consider an example with $p = 2$ and $K = 3$ with $\pi_k = 1/3$:





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